

Exercise 2.7 (1-3, 9-10)

1. $\forall x \in \mathbb{R}, x^2 > 0$
For every real number x , $x^2 > 0$
This is false. Take $x = 0 (0 \in \mathbb{R})$, then $x^2 = 0^2 = 0$ which is not greater than 0
2. $\forall x \in \mathbb{R}, \exists n \in \mathbb{N}, x^n \geq 0$
For every real number x , there exists some n in the natural numbers such that $x^n \geq 0$
True. Take $n = 2$, then $x^n = x^2 \geq 0$
3. $\exists a \in \mathbb{R}, \forall x \in \mathbb{R}, ax = x$
There exists some real number a in which, for all real numbers x , $ax = x$
True. Take $a = 1 \Rightarrow ax = 1x = x$
9. $\forall n \in \mathbb{Z}, \exists m \in \mathbb{Z}, m = n + 5$
For every integer n , there is some integer m such that $m = n + 5$
True. Taking $m = n + 5$ proves this is valid for every $n \in \mathbb{Z}$
10. $\exists m \in \mathbb{Z}, \forall n \in \mathbb{Z}, m = n + 5$
There is some integer m such that, for every integer n , $m = n + 5$
False. If $m = n + 5$, then for $n + 1 \in \mathbb{Z}, m = n + 5 \neq (n + 1) + 5$
An example could be: $n = 2 \Rightarrow m = 7$, but for $n = 3, n + 5 = 8 \neq 7$